17 a, b, c are positive integers such that $a > b > c$

N is the largest three digit number that has the digits a, b and c.
K is the smallest three digit number that has the digits a, b and c.

(a) Use algebra to show that the difference between N and K is always a multiple of 99

Largest $N = 100a + 10b + c$.

1 mark
Place-value expression

Smallest $K = 100c + 10b + a$.

$N - K = 99a - 99c = 99(a - c)$.

1 mark
Subtract expressions

$a - c$ is an integer, so $N - K$ is a multiple of 99.

1 mark
Integer factor proves it

(3)

(b) If $a > b$ and $b = c$ will the difference between N and K still be a multiple of 99?
Justify your answer.

Yes. The two $10b$ terms still cancel.

1 mark
Middle digit cancels

The difference is still $99(a - c)$, even if $b = c$.

(1)

(Total for Question 17 is 4 marks)